Chapter 2 - The Real Number System

Note 1

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"The real number system is the foundation upon which all of modern analysis rests."

- Walter Rudin, Principles of Mathematical Analysis (paraphrased essence) -

In this chapter, we will discuss the essential properties of the real number system $\mathbb{R}$. Although it is possible to give a formal construction of this system on the basis of a more primitive set (such as the set $\mathbb{N}$ of natural numbers or the set $\mathbb{Q}$ of rational numbers), we have chosen not to do so. Instead, we exhibit a list of fundamental properties associated with the real numbers and show how further properties can be deduced from them. This kind of activity is much more useful in learning the tools of analysis than examining the logical difficulties of constructing a model for $\mathbb{R}$.

2.1 : Preliminaries

Every rigorous study of mathematics begins with certain undefined concepts, elementary notions on which the theory is based, and certain postulates, properties which are assumed to be true and given no proof. Our study will be based on the primitive notions of real numbers and sets, which will be discussed in this section.

- We shall use standard notation for sets and real numbers. For example, $\mathbb{R}$ or $(-\infty, \infty)$ represents the set of real numbers.
- $\emptyset$ represents the empty set (the set with no elements).
- $a \in A$ means that a is an element of A, and $a \notin A$ means that a is not an element of A.
- We can represent a given finite set in two ways. We can list its elements directly, or we can describe it using sentences or equations. For example, the set of solutions to the equation $x^2 = 1$ can be written as $\{1, -1\}$ or $\{x : x^2 = 1\}$.
- A set A is said to be a subset of a set B (notation: $A \subseteq B$) if and only if every element of A is also an element of B. If A is a subset of B but there is at least one element $b \in B$ that does not belong to A, we shall call A a *proper subset* of B (notation $A \subset B$).

- Two sets A and B are said to be equal (notation: $A = B$) if and only if $A \subseteq B$ and $B \subseteq A$. If A and B are *not* equal, we write $A \neq B$.

- A set A is said to be *non-empty* if and only if $A \neq \emptyset$.

- The union of two sets A and B (notation: $A \cup B$) is the set of elements x such that x belongs to A or B or both.

- The intersection of two sets A and B (notation: $A \cap B$) is the set of elements x such that x belongs to both $A \cup B$.

- The complement of B relative to A (notation: $A \sim B$, sometimes B^c if A is understood) is the set of elements x such that $x \in A$ but $x \notin B$.

- To illustrate the above set operations, consider the following example:

$$\{-1, 0, 1\} \cup \{1, 2\} = \{-1, 0, 1, 2\}, \quad \{-1, 0, 1\} \cap \{1, 2\} = \{1\}$$
$$\{1, 2\} \sim \{-1, 0, 1\} = \{2\} \text{ and } \{-1, 0, 1\} \sim \{1, 2\} = \{-1, 0\} .$$

- Let X and Y be sets. The Cartesian product of X and Y is the set of ordered pairs defined by

$$X \times Y := \{(x, y) : x \in X, y \in Y\} .$$

  (The symbol := means "**equal by definition**" or "**is defined to be**".)

- Two points $(x, y), (z, w) \in X \times Y$ are said to be equal if and only if $x = z$ and $y = w$.

- Let X and Y be sets. A relation on $X \times Y$ is any subset of $X \times Y$. Let $\mathcal{R}$ be a relation on $X \times Y$. The domain of $\mathcal{R}$ is the collection of $x \in X$ such that (x, y) belongs to $\mathcal{R}$ for some $y \in Y$. The range of $\mathcal{R}$ is the collection of $y \in Y$ such that (x, y) belongs to $\mathcal{R}$ for some $x \in X$. When $(x, y) \in \mathcal{R}$, we shall frequently write $x\mathcal{R}y$.

- A function f from X into Y (notation: $f : X \to Y$) is a relation on $X \times Y$ whose domain is X (notation: $\text{Dom}(f) := X$) such that for each $x \in X$ there is a unique (one and only one) $y \in Y$ that satisfies $(x, y) \in f$. If $(x, y) \in f$, we shall call y the value of f at x (notation: $y = f(x)$ or $f : x \mapsto y$) and call x a pre-image of y under f. We said a pre-image because, in general, a point in the range of f might have more than one pre-image. For example, since $\sin(k\pi) = 0$ for $k = 0, \pm 1, \pm 2, \ldots$, the value 0 has infinitely many pre-images under $f(x) = \sin(x)$.

- If f is a function from X into Y, we will say that f is defined on X and call Y the codomain of f. The range of f is the collection of all values of f; that is, the set $\text{Ran}(f) := \{y \in Y : f(x) = y \text{ for some } x \in X\}$. In general, then, the range of a function is a subset of its codomain and each $y \in \text{Ran}(f)$ has one or more pre-images. If $\text{Ran}(f) = Y$ and each $y \in Y$ has exactly one pre-image, $x \in X$, under f, then we shall say that $f : X \to Y$ has an inverse, and shall define the inverse function $f^{-1} : Y \to X$ by $f^{-1}(y) := x$, where $x \in X$ satisfies $f(x) = y$.

- At this point it is important to notice a consequence of defining a function to be a set of ordered pairs. By the definition of equality of ordered pairs, two functions f and g are equal if and only if they have the same domain, and same values; that is, $f, g : X \to Y$, and $f(x) = g(x)$ for all $x \in X$. If they have different domains, they are different functions.

  - For example, let $f(x) = g(x) = x^2$.
  - Then $f : [0, 1) \to [0, 1)$ and $g : (-1, 1) \to [0, 1)$ are two different functions.
  - They both have the same range, $[0, 1)$, but each $y \in (0, 1)$ has exactly one pre-image under f, namely $\sqrt{y}$, and two pre-images under g, namely $\pm\sqrt{y}$.
  - In particular, f has an inverse function, $f^{-1}(x) = \sqrt{x}$, but g does not.

  Making distinctions like this will actually make our life easier later in the course.

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